Mabel Q. Yao

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PROFESSIONAL SUMMARY

Mabel Yao is a researcher specializing in Statistical Machine Learning, Statistical Inference, Computational Statistics, Explainable AI, and Data Science. She earned a Ph.D. in Statistics and an M.S. in Computer Science from North Dakota State University. Her academic background combines statistics, computing, and engineering, enabling her to develop data-driven solutions for real-world applications.

FIELDS OF SPECIALIZATION

- Statistical Machine Learning
- Statistical Inference, Computational Statistics
- Explainable AI (XAI)
- Network & Graph Data Analysis
- Data Science

EDUCATION

Statistics, Data Science, Computer Science

- **Ph.D. in Statistics**, North Dakota State University, USA — 2022.05-2026.05
  GPA: 4.0/4.0
- **M.S. in Computer Science**, North Dakota State University, USA — 2019.01-2022.05
  GPA: 4.0/4.0
- **B.S. in Software Engineering**, Dalian Jiaotong University, China — 2009.09-2013.07
  GPA: 87/100

Engineering

- **M.S. in Structural Engineering**, Tohoku University, Japan — 2013.09-2016.03
  GPA: 3.7/4.0
- **UC Davis Extension**, University of California, Davis, USA — 2015.02-2015.03
- **B.S. in Civil Engineering**, Dalian Jiaotong University, China — 2009.09-2013.07
  GPA: 87/100

WORKING EXPERIENCES

- **Freelance | Independent Researcher** — 2026.05–present
  AI & Statistics, Statistical Machine Learning, Data Science
  - Conduct independent research in statistical machine learning, explainable AI, and data science with emphasis on publishable methodologies and real-world applications.
- **Teaching Assistant in Statistics & Computer Science** — 2019.01–2026.05
  North Dakota State University

- Assisted instruction for 12 Statistics and Computer Science courses by leading weekly review sessions, creating practice problems in Python and Minitab, and answering student questions, which supported student learning outcomes and increased course completion rates.
- Developed clear lecture notes and lab exercises using LaTeX and Microsoft Office Suite, created sample code in Java and Python, provided detailed assignment feedback, and tutored students to boost understanding of statistics and programming, leading to improved class performance.
- Analyzed student performance data with R (using tidyverse and ggplot2) to identify trends and gaps, then presented findings to faculty, which led to targeted tutoring interventions that improved student pass rates.

• **Research Assistant in Data Analysis for Plant Sciences** 2019.01–2023.08
North Dakota State University

- Led design, execution, and analysis of agricultural experiments, using SAS for data analysis and Tableau for visualizations, and presented key findings that improved crop yield predictions to stakeholders.

• **Early Career Engineering Positions** 2014–2018

Structural Engineer, *Shenzhen Yuanlizhu Engineering Consultants Co.,Ltd* 2017.01-2018.12

- Utilized computer-aided engineering (CAE) tools to conduct structural design and improved structural performance, accuracy, and project efficiency; Collaborated with clients, investors, contractors, and multidisciplinary design teams to optimize structural solutions, streamline project execution, and support successful project delivery within technical and business requirements.

Project Assistant, *Shanghai Saiyo Construction Technology Co.,Ltd* 2016.09-2016.12

- Contributed to a Japanese shopping mall construction project in Ningbo by applying Building Information Modeling (BIM) to develop detailed virtual building models, enabling early-stage design validation and clash detection; Improved coordination among stakeholders enhanced project efficiency, and supported timely project execution.

Intern, *Yamashita Sekkei INC. Tohoku Branch* 2015.09-2015.10

- Performed structural analysis incorporating seismic isolators and seismic control devices to enhance structural stability, safety, and seismic performance; Produced detailed construction drawings using AutoCAD, contributing to accurate technical documentation and efficient coordination between design and construction teams.

Research Assistant in Structural Engineering, *Tohoku University* 2014.03–2016.03

- Assisted in seismic evaluation experiments for building structures by supporting experimental design, test implementation, and data collection processes; Contributed to the analysis of structural performance under seismic conditions through accurate experimental execution and systematic data acquisition, helping support engineering assessments and research outcomes.

PROFESIONAL ACTIVITIES

• **Statistics & Data Science Conferences** (2023–2025)

- **Red River Valley Statistics Conference** Presented: Testing Common Invariant Subspace of Multilayer Networks; Analysis of Time Series Models for Electricity Consumption Forecasting; Survival Analysis for Colorectal Cancer Considering Effects of Staging and Treatment Variables; Statistical Learning for Virtual Screening in Drug Discovery.

• **Engineering Conferences** (2014–2015)

- **Qianqian Yao**, et al. "Effect of Earthquake Response Spectrum Characteristics on Residual

Seismic Performance", Japan Concrete Institute, vol.37 NO.2, 685-690. 2015.

- **Qianqian Yao**, et al. "Seismic Evaluation of Damaged RC buildings Basing on Earthquake Response Spectrum, part 1 Basic Evaluation Basing on Seismic Capacity Index and Effect of Building Capacity Reduction", published in the Architectural Institute of Japan meeting in Koube, page195-196, 2014.8.

- **Scholarship & Assistantship**

- Graduate Teaching Assistantship at North Dakota State University, 2019.01-2026.05
- Inter-Graduate School Doctoral Degree Program on Science for Global Safety at Tohoku University, 2014.03-2016.03

- **Competition**

- Intelligent Ground Vehicle Competition (Autonav Challenge, 2019)

PUBLICATIONS

- **Qianqian Yao**. Empirical Likelihood Test for Common Invariant Subspace of Multilayer Networks based on Monte Carlo Approximation. (2026) *arXiv:2601.06390*
- Mingao Yuan and **Qianqian Yao**. Testing common degree-correction parameters of multilayer networks. *Statistics and Computing*, 35(2), 38. 2025.

THESES

- **Qianqian Yao**. Testing Common Invariant Subspace of Multilayer Networks. Doctoral Dissertation at NDSU. (2026)
- **Qianqian Yao**. Comparison of Non-Learned and Learned Molecule Representations for Catalyst Discovery. Master Thesis at NDSU. (2022)

TECHNICAL SKILLS

- **Programming:** Python, R, SQL.
- **Machine Learning:** scikit-learn, XGBoost, CatBoost, PyTorch.
- **Statistics:** Regression, GLM, Bayesian, Survival, Time Series.
- **Visualization:** Power BI, Tableau, matplotlib, ggplot.
- **Tools:** Git, Jupyter, LaTeX, Octave, Minitab, JMP, etc.

SELECTED DATA SCIENCE PROJECTS

- **Expected Credit Loss Prediction using Machine Learning**

  GitHub: Developed an end-to-end Expected Credit Loss framework integrating ML-based PD, LGD, and EAD models using XGBoost, logistic regression, and explainability techniques, enabling portfolio-level credit loss forecasting under macroeconomic stress scenarios.

- **GLM for Credit Card Expenditure User Behavior Analysis**

GitHub: Developed and evaluated GLM-based models (Binomial, Poisson, Log-normal, Gamma, Quasibinomial); applied goodness-of-fit diagnostics, quasi-GLM variance functions, and weighted least squares (WLS) to model heteroskedasticity.

- **Time Series Analysis for Electricity Consumption Forecasting**

  GitHub: Implemented AR, MA, and ARIMA time-series models and conducted model comparison based on forecasting performance, identifying ARIMA as the best-performing model.

- **Survival Analysis for Colorectal Cancer considering Effects of Staging and Treatment**

  GitHub: Retrieved clinical data from the SEER*Stat database and performed survival analysis using Kaplan–Meier estimators and log-rank tests to compare treatment groups; Implemented Cox proportional hazards regression to evaluate the significance of treatment covariates and quantify their impact on survival outcomes.

- **EHR-based 30-Day Hospital Readmission Risk Modeling**

  GitHub: Built an end-to-end predictive modeling pipeline: data preprocessing, label engineering, patient-level train/test split, and model development using Logistic Regression, Random Forest, XGBoost, and CatBoost; conducted model comparison and SHAP interpretation to support risk stratification incorporating SDOH variables.